

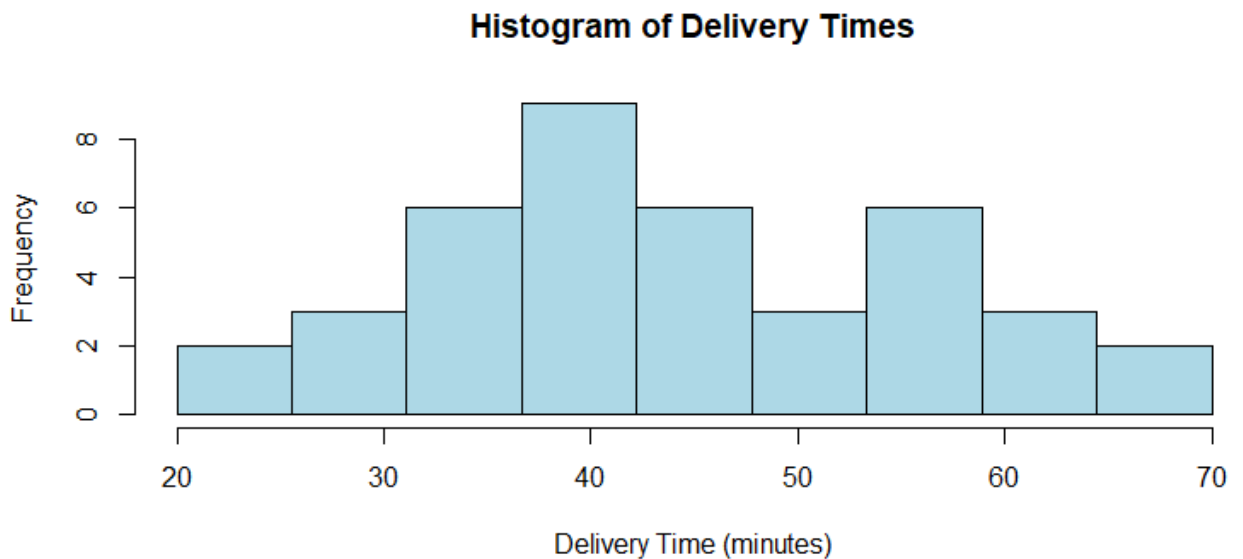
PS Lab 04

1. Import the dataset ('Exercise – Lab 05.txt') into R and store it in a data frame called "Delivery Times".

```
Delivery_Times <- read.table("Exercise - Lab 05.txt", header = TRUE)
names(Delivery_Times) <- c("x1")
```

2. Draw a histogram for deliver times using nine class intervals where the lower limit is 20 and upper limit is 70. Use right open intervals.

```
Delivery_Times <- read.table("Exercise - Lab 05.txt", header = TRUE)
names(Delivery_Times) <- c("x1")
breaks <- seq(20, 70, length.out = 10)
hist(Delivery_Times$x1,
     main = "Histogram of Delivery Times",
     xlab = "Delivery Time (minutes)",
     breaks = breaks,
     right = FALSE,
     col = "lightblue",
     border = "black")
```



3. Comment on the shape of the distribution.
 - the data seems evenly spread.

4. Draw a cumulative frequency polygon (ogive) for the data in a separate plot.

```
hist(Delivery_Times$x1,  
     main = "Histogram of Delivery Times",  
     xlab = "Delivery Time (minutes)",  
     breaks = breaks,  
     right = FALSE,  
     col = "lightblue",  
     border = "black")  
hist_data <- hist(Delivery_Times$x1,  
                  breaks = breaks,  
                  right = FALSE,  
                  plot = FALSE)  
cum_freq <- cumsum(hist_data$counts)  
plot(hist_data$breaks[-1], cum_freq,  
     type = "o",  
     main = "Cumulative Frequency Polygon (Ogive)",  
     xlab = "Delivery Time (minutes)",  
     ylab = "Cumulative Frequency",  
     col = "blue",  
     pch = 16)
```

